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The Shape of Evidence

Fuzzy Prediction Sets for Downstream Decisions

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Abstract

Sound statistical evidence does not guarantee good decisions. Consider a decision-maker whose loss depends on which outcome occurs. They ask a statistician for a report that scores each outcome by how strongly the data count against it, and the report must be valid, misleading no more often than a pre-agreed rate. Many reports may be valid, yet recommend different actions. Existing theory picks the report that accumulates evidence fastest, without considering the decision-maker's losses at stake. This thesis instead asks which report to choose when such loss function is known in advance.

Each report bounds an action's expected loss. The "safe" report minimises this bound by scoring outcomes in proportion to the action's losses, and leads to the action it "protects". By contrast, the report that accumulates evidence fastest may recommend a different action, in which case the protected action must be imposed as a constraint. That report scores outcomes by how much likelier the data make them, and recovers the safe report only when losses follow those likelihoods.

For a shopkeeper ordering shirts before the 2026 World Cup final, with Polymarket prices as the forecast, no likelihood-based report found in two searches selects the order the forecast recommends, unlike the safe report. In controlled simulations, the likelihood-based report picks the wrong action even when the forecast is right. Losses belong in the design of evidence, not its interpretation.

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Contents

1 Introduction **2**

1.1 From Prediction Sets to Scores 2

1.2 Two Ways to Judge a Profile 3

1.3 Contribution 4

2 Theoretical Framework **5**

2.1 The Decision Problem 5

2.2 Evidence Profiles and Validity 5

2.3 From Prediction Sets to the Weighted Rule 6

2.4 A Worked Example 7

3 Designing Evidence for a Decision **9**

3.1 Result 1: the Safest Profile for an Action 9

3.2 Result 2: Informing Is Not Inducing 11

3.3 The Protection–Information Frontier 14

3.4 Result 3: When Shape Functions Suffice 16

3.5 The Procedure, Worked Through 17

4 Application: Stocking Shirts for the World Cup Final **19**

4.1 Two Contracts, Four Outcomes 19

4.2 The Model and the Belief 21

4.3 The Two Criteria Disagree 22

4.4 No Increasing Shape Reaches the Target 23

5 Controlled Simulations: Scoring the Decisions Against a Known Truth . **25**

5.1 The Likelihood Ratio Moves Against the Belief 25

5.2 How Wide the Disagreement Is 26

5.3 How Far an Optimistic Belief Can Be Trusted 27

6 Conclusion **29**

6.1 What the Thesis Shows 29

6.2 Limitations 29

6.3 Open Questions 29

References **30**

A Notation **31**

1 Introduction

1.1 From Prediction Sets to Scores

Consider a decision-maker who must act before an uncertain outcome is known. They know which outcomes may occur, which actions are available to take, and what each action would cost in each outcome. What they do not know is which outcome will occur, so they ask a statistician for evidence.

Call whatever the statistician sends back a report. It might be a shortlist of the outcomes still worth taking seriously, or a score for each outcome saying how strongly the data count against it. Whatever its form, a report must be valid, meaning it must not mislead more often than a pre-agreed error rate allows.

The established way to build such a report is to build a prediction set, a shortlist of the outcomes that remain plausible. Conformal prediction constructs such sets with a coverage guarantee: for a chosen level $\alpha \in (0, 1)$, the shortlist S contains the realised outcome Z with probability at least $1 - \alpha$, that is, $P[Z \in S] \geq 1 - \alpha$ [Vovk et al., 2005, Angelopoulos and Bates, 2023]. A cautious decision-maker can then protect against the worst thing the shortlist still permits,

$$\delta_S \in \arg\min_{d \in \mathcal{D}} \max_{z \in S} L(d, z), \tag{1.1}$$

writing $L(d, z)$ for what action d costs when outcome z occurs. Whenever the realised outcome lies in S, the loss of δ_S cannot exceed the worst loss over S, so coverage carries over to the decision,

$$P\Big[L(\delta_S, Z) \leq \max_{z \in S} L(\delta_S, z)\Big] \geq 1 - \alpha. \tag{1.2}$$

This is the basis of certified-decision methods [Andrews and Chen, 2025, Kiyani et al., 2025].

This guarantee is genuine, but a shortlist records only membership. Once two outcomes are both in it, the set no longer says which one the data support more strongly, so the rule treats a barely plausible outcome and a highly likely one identically. The recommended action can also change with the coverage level even though nothing about the decision has changed.

E-values keep what a shortlist discards. Rather than sorting each outcome in or out, they give every outcome its own score $\mathcal{E}(z) \geq 0$, large where the data argue against that outcome and small where they do not. The collection of these scores, one per outcome, is what the statistician hands over. This thesis calls it a profile and writes it $\mathcal{E}$.

What stops a profile from claiming strong evidence everywhere is a single requirement,

$$\mathbb{E}_P[\mathcal{E}(Z)] \leq 1, \tag{1.3}$$

that the scores average to at most one under the distribution P the evidence is weighed against. Section 2 extends this to a whole set $\mathcal{P}$ of distributions.

A profile satisfying this restriction is valid, and a valid profile is also called an e-variable. The condition acts as a budget: raising the score on one outcome forces a reduction elsewhere.

In return, the scores can be compared against any threshold, even one chosen after they are seen, without losing the guarantee (Section 2.2). This is the basis of safe testing [Shafer, 2021, Grünwald et al., 2024, Ramdas et al., 2023]. Fuzzy prediction sets carry this graded evidence back to the prediction problem, replacing the in-or-out verdict of a shortlist with a score for every outcome [Koning and van Meer, 2025, Vovk, 2025, Gauthier et al., 2025].

1.2 Two Ways to Judge a Profile

Validity rules out overclaiming, but it leaves a great deal open. Many profiles satisfy it, and they need not lead the decision-maker to the same action. The statistician must still choose one, and could let what each mistake would cost guide that choice. This is where the existing literature and the decision-maker part company.

The standard construction compares two distributions. One is P, the distribution validity is required to hold under. The other is the belief $\widehat{P}_1$, what the decision-maker expects to happen. Their ratio

$$r(z) := \frac{\widehat{P}_1(z)}{P(z)} \tag{1.4}$$

is large where the belief rates an outcome likelier than P does, and the profile is built by passing this ratio through an increasing shape function w, reporting $\mathcal{E}(z) = w(r(z))$.

The usual measure of how much a profile tells us is its expected log-evidence under the belief, $\mathbb{E}_{\widehat{P}_1}[\log \mathcal{E}(Z)]$, which is high when large scores sit where the belief puts its probability. Whenever P gives every outcome positive probability, $w(r) = r$ maximises it among valid profiles [Grünwald et al., 2024, Koning, 2025]. Statisticians with other preferences over how evidence accumulates choose other shape functions, and Koning and van Meer [2025] determine which w is best once such a preference is fixed.

However, all of this concerns how much the evidence tells us. It says nothing about what happens when the evidence must produce an action. For that, Koning and van Meer [2025] score each action by

$$C(d; \mathcal{E}) := \max_{z \in \mathcal{Z}} \frac{L(d, z)}{\mathcal{E}(z)}, \tag{1.5}$$

its largest loss across outcomes, with each loss first divided by the evidence against the outcome that would produce it. Validity turns this single number into a bound on the action's expected loss, and the decision-maker takes whichever action has the smallest one.

Dividing by the score is how evidence lowers the bound. A large score on an outcome shrinks the loss the action would suffer there, while a small score leaves that loss large. Because of the budget, the scores cannot all be large, so a profile has to choose where its large scores go. The bound is the worst loss after division, so it stays low only if the large scores sit where the action's losses are large.

This is where the standard construction can go wrong. It puts its large scores on the outcomes the belief expects, those that $\widehat{P}_1$ rates likelier than P does, and is therefore left with small scores on the outcomes the belief considers unlikely. If the action is expensive

in one of those unlikely outcomes, that loss is not reduced, and the action's bound is poor. Which outcomes the belief expects and where the decision-maker has most to lose are separate questions, and there is no reason their answers should coincide.

An evidence profile can thus be judged in two ways, by the information it preserves, or by the protection it gives to a decision. The two judgements need not agree, which leads to the research question of the thesis:

> Given a loss function and a stated attitude toward risk, how should a statistician choose the e-variable supplied to a decision-maker?

1.3 Contribution

Existing theory chooses a profile by a preference over evidence, and cannot say which preference represents a given decision-maker's loss. The closest predecessor is the e-posterior of Grünwald [2023], which supports frequentist decision-making without a prior and already divides a loss by an e-value in the manner of (1.5). Neither supplies a design step in which the loss function is an input to the construction of the profile itself. This step is the contribution of this thesis, which uses three results to answer the question.

The first asks how low the bound on a chosen action's loss can be pushed. The answer is to make the scores proportional to what that action costs in each outcome, and the bound then equals that action's largest expected loss across the distributions validity must hold under (Theorem 3.1). No other valid profile does as well, and the optimum is unique. Call it the safe profile. It is built from the loss and those distributions alone, while the belief plays no part in it.

The second result separates informing from inducing. The safe profile always leads to the action it protects, unless some other action is cheaper in every outcome, but the most informative profile, the likelihood ratio, need not, even when the target is the action the belief itself favours. Keeping as much information as possible while still reaching the target therefore requires imposing the target as a constraint, and the optimum then sits where the target ties with a competitor (Proposition 3.3).

The third asks whether the standard construction could have found the safe profile anyway. Since it reads each outcome only through r, two outcomes with the same likelihood ratio must receive the same score, however differently they cost. The safe profile is built from the loss and can separate them. The two agree exactly when the loss is determined by the ratio (Proposition 3.4). Otherwise, no shape function recovers the safe profile, so the loss must enter the design of the profile rather than be applied to its output.

2 Theoretical Framework

2.1 The Decision Problem

Let Z be an unknown outcome in a finite outcome space $\mathcal{Z}$, and let $d \in \mathcal{D}$ be an action chosen before Z is observed. The loss $L(d, z) \geq 0$ is what choosing d costs when the outcome turns out to be z. The decision-maker supplies this loss table, and it records which mistakes matter and by how much. With finitely many outcomes the table is a grid of numbers, one per action and outcome, and $L(d, \cdot)$ denotes the row for action d, listing what that action costs in each outcome in turn.

Uncertainty about outcome Z enters in two separate places. The first is the model $\mathcal{P}$, a set of distributions any one of which could be the true one. The decision-maker is unwilling to stake the guarantee on a single distribution, so the report is required to be valid under every member at once. As $\mathcal{P}$ gets larger, the statement about what the world might be doing becomes more cautious, and it buys a weaker guarantee in return.

The second is the belief $\widehat{P}_1$, the single distribution the decision-maker actually thinks is true, formed before any data is collected. It carries no guarantee whatever. Intuitively, it is simply what they would act on "if nothing could go wrong".

The two do different jobs. The model says what the report must survive, the belief says what it should be good at. They are not competing estimates of one quantity, and the thesis is about what happens when they disagree about which action to take. Two standing assumptions apply throughout: every distribution in $\mathcal{P}$ gives every outcome positive probability, so each expected loss is a finite sum, and every expectation is taken under the distribution named in its subscript.

2.2 Evidence Profiles and Validity

A profile $\mathcal{E}$ is a valid e-variable for the model $\mathcal{P}$ when

$$\mathbb{E}_P[\mathcal{E}(Z)] \leq 1 \qquad \text{for every } P \in \mathcal{P}. \tag{2.1}$$

Read as a budget, the condition gives the profile one unit of e-value to spend on average. Nothing dictates how that unit is spread across outcomes, and how to spread it is precisely the design problem of this thesis. Constructing a valid profile from raw data is a separate problem, solved by the methods cited in Section 1, and is taken as given here.

What makes the condition worth accepting is what it buys. Markov's inequality turns it into

$$P[\mathcal{E}(Z) \geq 1/\alpha] \leq \alpha \qquad \text{for every } \alpha \in (0, 1), \tag{2.2}$$

so a profile whose score at an outcome exceeds $1/\alpha$ can be treated as evidence at level α against it. This holds at every α simultaneously and under every member of the model, so a single profile serves every threshold a reader might apply with no correction for inspecting several. A companion result [Grünwald, 2024, Koning, 2024] extends it to thresholds chosen

after the scores are seen, so evidence can be published before anyone fixes how strictly to read it.

One further consequence matters below. Only the hardest constraint in (2.1) does any work: the profile is valid exactly when its largest expectation across the model is at most one, so validity is settled by the single member under which the profile looks largest. A profile that spends the budget exactly under some member, rather than leaving slack everywhere, is the interesting case, since any leftover could be spent on sharper evidence. Every profile constructed in this thesis is of that kind. The examples below use the simplest model, a single distribution $\mathcal{P} = \{P_0\}$, to keep the arithmetic checkable, and Section 4 uses a genuinely composite one.

2.3 From Prediction Sets to the Weighted Rule

A profile has to become an action. For a prediction set this is done by the rule (1.1), which protects against the worst outcome the set still admits. The profile analogue replaces the worst case over a set by a worst case weighted by the evidence. This is the weighted worst-case rule of Koning and van Meer [2025],

$$\delta(\mathcal{E}) \in \arg\min_{d \in \mathcal{D}} C(d; \mathcal{E}), \qquad C(d; \mathcal{E}) := \max_{z \in \mathcal{Z}} \frac{L(d, z)}{\mathcal{E}(z)}, \tag{2.3}$$

with the conventions $x/0 = \infty$ for $x > 0$ and $0/0 = 0$. Dividing by the score does not declare an outcome impossible: it discounts a loss that would be incurred only if the outcome occurred, in proportion to the evidence against it.

The number $C(d; \mathcal{E})$ is called the certificate of d, because it bounds expected loss. It depends only on the loss row and the profile, so it is known before the outcome is. Assume $C(d; \mathcal{E}) < \infty$, since otherwise the bound is vacuous. By definition of the maximum, $L(d, z) \leq C(d; \mathcal{E})\, \mathcal{E}(z)$ at every outcome, so for any $P \in \mathcal{P}$,

$$\mathbb{E}_P[L(d, Z)] \leq C(d; \mathcal{E})\, \mathbb{E}_P[\mathcal{E}(Z)] \leq C(d; \mathcal{E}). \tag{2.4}$$

The final inequality is where validity is used, and without it a profile could be scaled up to make any certificate small. The bound holds under every member of the model at once, and it concerns expected loss, not every realised loss. The same pointwise inequality also gives a tail bound. A loss of at least $C(d; \mathcal{E})/\alpha$ forces a score of at least $1/\alpha$, so by (2.2)

$$P\big[L(d, Z) \geq C(d; \mathcal{E})/\alpha\big] \leq \alpha \qquad \text{for every } \alpha \in (0, 1) \text{ and every } P \in \mathcal{P}.$$

The rule may have several minimisers. An action is weakly selected when it is one of them, and strictly selected when its certificate beats every competitor. That distinction matters once an action is imposed as a constraint in Section 3.2.

2.4 A Worked Example

A bettor stakes money on a football match with two outcomes, a win for A or a win for B. They may back A through d_1 or hedge the position through d_2. A losing bet costs twice as much as a hedge that turns out to have been unnecessary. Listing each action's loss if A wins and then if B wins,

$$L(d_1, \cdot) = (0, 8), \qquad L(d_2, \cdot) = (4, 0), \tag{2.5}$$

so backing A loses 8 only if B wins, and the hedge loses 4 only if A wins. The model is the single distribution $\mathcal{P} = \{P_0\}$ with $P_0 = (1/2, 1/2)$, an even match, and the bettor's belief is $\widehat{P}_1 = (0.75, 0.25)$, favouring A.

The two criteria already disagree about what to do. The certificate bounds expected loss under the model (2.4), so a decision-maker who wants that guarantee compares actions by their average loss against P_0,

$$\mathbb{E}_{P_0}[L(d_1, Z)] = \tfrac{1}{2}(0) + \tfrac{1}{2}(8) = 4,$$
$$\mathbb{E}_{P_0}[L(d_2, Z)] = \tfrac{1}{2}(4) + \tfrac{1}{2}(0) = 2,$$

and the hedge d_2 comes out ahead. A decision-maker who trusts the belief averages under $\widehat{P}_1$ instead,

$$\mathbb{E}_{\widehat{P}_1}[L(d_1, Z)] = 0.75(0) + 0.25(8) = 2,$$
$$\mathbb{E}_{\widehat{P}_1}[L(d_2, Z)] = 0.75(4) + 0.25(0) = 3,$$

so the belief favours backing A. Neither answer is wrong. They answer different questions, and the decision-maker has to say which one they are asking.

Now compare the two decision rules on this example. A profile becomes a prediction set by keeping every outcome whose score stays below $1/\alpha$, since by (2.2) a larger score counts as evidence at level α against it. Fix $\alpha = 1/2$, so the threshold is 2, and take two valid profiles: the flat profile $\mathcal{E}^{\text{flat}} = (1, 1)$, which spends its budget evenly, and a graded profile $\mathcal{E}^{\text{graded}} = (2/3, 4/3)$, which shifts weight onto B. Both are valid, since each averages to one under P_0.

Neither profile reaches the threshold at either outcome, so both produce the same prediction set $S = \{A, B\}$, and (1.1) cannot tell them apart. In both cases the worst loss over S is 8 for the bet and 4 for the hedge, so each backs the hedge.

The weighted rule separates them, because it divides outcome by outcome before taking the worst case. Under the flat profile,

$$C(d_1) = \max(0/1,\ 8/1) = 8, \qquad C(d_2) = \max(4/1,\ 0/1) = 4,$$

so d_2 wins outright, exactly as the prediction set concluded. The graded profile changes the divisors,

$$C(d_1) = \max(0/\tfrac{2}{3},\ 8/\tfrac{4}{3}) = 6, \qquad C(d_2) = \max(4/\tfrac{2}{3},\ 0/\tfrac{4}{3}) = 6,$$

and the two are now equal, so d_1 becomes available. The extra score on B deflates the only outcome in which backing A costs anything, while the score removed from A inflates the cost of hedging. The consequence is a different action, not merely a different number, and under the belief d_1 costs 2 where d_2 costs 3. The dependence on α disappears too, since (2.3) never thresholds the profile, so no coverage level enters the choice of action.

This is the whole problem in miniature: two valid profiles recommend different actions, and validity does not choose between them.

A second example shows that the disagreement can also turn on a single outcome that the belief considers rare but that is costly for most actions. Showing this needs a third outcome, a draw, which settles both side bets as losses at once. The bettor may back either side through d_A or d_B, or insure against the draw through d_C, which costs a premium whenever the match has a winner and pays only in the draw. Writing the outcomes as a win for A, a win for B, and a draw C,

$$L(d_A, \cdot) = (0, 2, 10), \quad L(d_B, \cdot) = (2, 0, 10), \quad L(d_C, \cdot) = (4, 4, 0), \tag{2.6}$$

with $\mathcal{P} = \{P_0\}$ for $P_0 = (1/3, 1/3, 1/3)$ and belief $\widehat{P}_1 = (0.6, 0.3, 0.1)$. The belief is confident about the side and dismissive about the draw, putting 0.60 on A but only 0.10 on C, which is plausibly too low for football, where draws are common.

The two criteria disagree here too. Against the model the averages are $(4, 4, 8/3)$, so protection points to d_C, while under the belief they are $(1.6, 2.2, 3.6)$, so the belief points to d_A. The draw decides it: d_A and d_B each expose the bettor to a loss of 10 there, which d_C avoids at the cost of never doing well. Section 5.3 asks how wrong such a belief can be before trusting it stops paying.

3 Designing Evidence for a Decision

The previous section left a problem. Many profiles are valid, and they can lead the decision-maker to different actions, so validity alone cannot tell the statistician which profile to report. Something else has to decide.

The decision-maker first has to name the action they want the profile to support, called the target. Which action they name reflects their attitude toward risk. A decision-maker who wants the strongest guarantee names the action with the smallest worst-case expected loss across the model, called the protective action. One who trusts the belief names the action with the smallest expected loss under the belief, i.e. the action that would cost least on average if the belief were true.

With the target named, the statistician designs the profile, and three results show how. The first finds the profile that gives the target the strongest guarantee. The second shows how to trade some of that guarantee for information without losing the target. The third shows that the standard construction, which reshapes the likelihood ratio, cannot in general find the first.

3.1 Result 1: the Safest Profile for an Action

Start with protection. Fix an action d. Its certificate $C(d;\mathcal{E})$ bounds the expected loss of d under every distribution in the model, so the lower the certificate, the stronger the guarantee. The question is how low a valid profile can push it.

The answer can be guessed. The certificate is the largest of the ratios $L(d,z)/\mathcal{E}(z)$, one for each outcome. To keep the largest one small, the profile should put large scores where the action is expensive. But the scores share a budget, so a large score on one outcome means a smaller score elsewhere. The best compromise is to make all the ratios equal, so that none is larger than it needs to be, and that means scores proportional to the losses. The theorem confirms this.

Theorem 3.1 (Certificate optimum)**.** *Fix an action $d \in \mathcal{D}$ and let*

$$\mu_d := \max_{P\in\mathcal{P}} \mathbb{E}_P[L(d,Z)]$$

be its worst expected loss across the model, assumed strictly positive and attained by some $P^\star \in \mathcal{P}$. Every valid e-variable $\mathcal{E}$ satisfies

$$C(d;\mathcal{E}) \geq \mu_d,$$

where $C(d;\mathcal{E})$ is the certificate (2.3), and equality is attained by

$$\mathcal{E}_d^{\text{safe}}(z) = \frac{L(d,z)}{\mu_d}.$$

Moreover this profile is the only one attaining the bound: any valid $\mathcal{E}$ with $C(d;\mathcal{E}) = \mu_d$ equals $\mathcal{E}_d^{\text{safe}}$ at every outcome.

Proof. Let $\mathcal{E}$ be a valid e-variable and write $C := C(d;\mathcal{E})$ for its certificate. If $C = \infty$ the bound is immediate, so assume $C < \infty$. Since C is the largest of the ratios $L(d,z)/\mathcal{E}(z)$, every outcome satisfies $L(d,z) \le C\,\mathcal{E}(z)$. Taking expectations of both sides under $P^\star$ gives

$$\mu_d = \mathbb{E}_{P^\star}[L(d,Z)] \le C\,\mathbb{E}_{P^\star}[\mathcal{E}(Z)] \le C,$$

where the final inequality is validity of $\mathcal{E}$ under $P^\star$. This proves the lower bound.

For attainment, $\mathcal{E}_d^{\text{safe}}$ is non-negative and valid, because for every $P \in \mathcal{P}$

$$\mathbb{E}_P[\mathcal{E}_d^{\text{safe}}] = \frac{\mathbb{E}_P[L(d,Z)]}{\mu_d} \le 1,$$

the inequality holding because μ_d is the largest such expected loss, with equality at $P = P^\star$. The safe profile is therefore exact under $P^\star$, spending its whole budget there. Where $L(d,z) > 0$ the ratio $L(d,z)/\mathcal{E}_d^{\text{safe}}(z)$ equals μ_d, and where $L(d,z) = 0$ it equals zero under the convention $0/0 = 0$. Since $\mu_d > 0$, at least one outcome has $L(d,z) > 0$, so the largest ratio is μ_d.

For uniqueness, suppose a valid $\mathcal{E}$ attains $C = \mu_d$. Both inequalities in the displayed chain are then equalities. The second forces $\mathbb{E}_{P^\star}[\mathcal{E}] = 1$, so every optimal profile is exact under $P^\star$, and the first forces

$$\sum_{z\in\mathcal{Z}} P^\star(z)\big(\mu_d\,\mathcal{E}(z) - L(d,z)\big) = 0.$$

Every bracket is non-negative and every $P^\star(z)$ is strictly positive, so a sum of non-negative terms can vanish only if each term does. Hence $\mu_d\,\mathcal{E}(z) = L(d,z)$ at every outcome, which is $\mathcal{E} = L(d,\cdot)/\mu_d$. □

In words, the best guarantee any valid profile can give d is its worst-case expected loss μ_d, and only one profile achieves it, the one that scores each outcome in proportion to what d costs there. Call it the safe profile for d.

Three features of the safe profile matter below. First, only one member of the model is relevant. Validity holds with equality under $P^\star$, the distribution under which d looks worst, and with room to spare under members that make d look cheaper, so adding members to the model changes nothing unless one of them makes d look worse than $P^\star$ does.

Second, the best guarantee available for any action is

$$\min_{d\in\mathcal{D}} \max_{P\in\mathcal{P}} \mathbb{E}_P[L(d,Z)],$$

which the protective action attains. Third, the safe profile never uses the belief. It is built from the loss and the model alone, so a decision-maker who wants only the strongest guarantee

has no use for the belief. That is not a defect of the belief, but what wanting only protection means.

A guarantee for an action is only useful if the decision-maker, handed the profile, actually takes that action. The rule compares all actions under the same profile, so a competitor could in principle come out cheaper. The next result shows that this happens only if the target was a poor choice to begin with.

Corollary 3.2 (The safe profile selects its action)**.** *Under the conditions of Theorem 3.1, $\mathcal{E}_d^{\text{safe}}$ weakly selects d, meaning d is among the actions the rule returns, unless some competitor d' costs less than d in every outcome where d costs anything and nothing where d costs nothing. In particular, it weakly selects any action that is cheapest against the model, and any action that is cheapest under a distribution giving every outcome positive probability.*

Proof. Under $\mathcal{E}_d^{\text{safe}}$ the target's certificate is μ_d, and a competitor's is

$$C(d';\mathcal{E}_d^{\text{safe}}) = \mu_d \max_{z\in\mathcal{Z}} \frac{L(d',z)}{L(d,z)},$$

with the conventions of (2.3). This falls below μ_d exactly when every ratio is below one, which is the stated condition. An action cheapest against the model cannot be beaten, since by Theorem 3.1 every competitor has $C(d';\mathcal{E}) \geq \mu_{d'} \geq \mu_d$. An action cheapest under a distribution Q with full support cannot be beaten either, since a competitor meeting the condition would have strictly smaller expected loss under Q. □

In other words, the safe profile can hand the decision only to an action that is cheaper than the target in every outcome, and no decision-maker would target an action like that. It may tie with a competitor, which is why selection is only weak.

3.2 Result 2: Informing Is Not Inducing

The safe profile gives the target the strongest guarantee, but it ignores the belief entirely. A decision-maker who also wants the evidence to reflect the belief needs something else. The standard answer is the likelihood ratio r of (1.4), the valid profile that maximises expected log-evidence under the belief [Grünwald et al., 2024, Koning, 2025]. The natural question is whether it leads to the target. It need not, even when the target is the action the belief itself recommends.

Take the two-outcome example, where the belief favours the bet d_1. There $P_0 = (1/2, 1/2)$ and $\widehat{P}_1 = (0.75, 0.25)$, so $r = (1.5, 0.5)$, and with $L(d_1,\cdot) = (0, 8)$ and $L(d_2,\cdot) = (4, 0)$ the certificates are

$$C(d_1;r) = \max(0/1.5,\ 8/0.5) = 16, \qquad C(d_2;r) = \max(4/1.5,\ 0/0.5) = 8/3.$$

The rule selects the hedge d_2. The reason is where the ratio puts its scores. It puts the large score on A, the outcome the belief expects, and the small score on B, the only outcome

in which the bet loses anything. Dividing the bet's loss of 8 by 0.5 doubles it instead of shrinking it. The profile that best reflects the belief leads the decision-maker away from the action the belief recommends.

So information and the target can conflict, and the target has to come first. The decision-maker names the target, written $d^\star$, and the statistician looks for the most informative profile among those that lead to it. A natural target is the action the belief recommends,

$$d^\star \in \arg\min_{d \in \mathcal{D}} \mathbb{E}_{\widehat{P}_1}[L(d, Z)],$$

though any target the decision-maker names is handled the same way. The target is fixed before the profile is designed, so it does not depend on $\mathcal{E}$. What depends on $\mathcal{E}$ is the action the rule returns, $\delta(\mathcal{E})$, and the requirement is that this action be $d^\star$.

Information is measured, as in Section 1.2, by expected log-evidence under the belief, $\mathbb{E}_{\widehat{P}_1}[\log \mathcal{E}(Z)]$. The problem is then

$$\begin{aligned} \max_{\mathcal{E}} \quad & \mathbb{E}_{\widehat{P}_1}[\log \mathcal{E}(Z)] \\ \text{subject to} \quad & \mathcal{E} \geq 0, \qquad \mathbb{E}_P[\mathcal{E}(Z)] \leq 1 \ \text{ for all } P \in \mathcal{P}, \\ & d^\star \in \arg\min_{d \in \mathcal{D}} C(d; \mathcal{E}). \end{aligned} \tag{3.1}$$

The first line of constraints says the profile is valid, and without the second line the answer would be the likelihood ratio. The second line is a constraint on the profile. It keeps only profiles under which the rule returns $d^\star$, however informative the others are, so it changes the answer only when the likelihood ratio leads elsewhere, as in the example.

What does the answer look like then? The next result shows that it sits exactly where the target stops winning.

Proposition 3.3 (Boundary structure)**.** *Assume in addition that $\mathcal{D}$ is finite, that $\widehat{P}_1(z) > 0$ for every outcome, so that expected log-evidence is finite inside the feasible set and strictly increasing in every coordinate, and that exactly one validity constraint binds at the optimum. Write $P^\star$ for the member it binds under, which is P_0 for a one-point model, and $r = \widehat{P}_1/P^\star$. Let*

$$F = \{\mathcal{E} \geq 0 : \ \mathbb{E}_P[\mathcal{E}] \leq 1 \text{ for all } P \in \mathcal{P}, \ d^\star \in \arg\min_{d} C(d; \mathcal{E})\}$$

be the feasible set of (3.1), and let $\mathcal{E}^\star$ be a maximiser of expected log-evidence over F with a finite objective. Then validity binds at $\mathcal{E}^\star$. If $d^\star$ is strictly selected at $\mathcal{E}^\star$, meaning its certificate is strictly smaller than every competitor's, then $\mathcal{E}^\star = r$. Consequently, if r does not select $d^\star$, the target ties with some competitor d' at the optimum, $C(d^\star; \mathcal{E}^\star) = C(d'; \mathcal{E}^\star)$, and is only weakly selected.

Proof. If every validity constraint is slack at $\mathcal{E}^\star$, scaling $\mathcal{E}^\star$ by a constant $c > 1$ raises the objective, which is strictly increasing in each coordinate, and scales every certificate by $1/c$, preserving the argmin. So at least one constraint binds. If $d^\star$ is strictly selected, every certificate comparison remains strict under sufficiently small perturbations of $\mathcal{E}^\star$, so $\mathcal{E}^\star$ is a local maximiser of expected log-evidence subject only to the binding validity constraint $\mathbb{E}_{P^\star}[\mathcal{E}] = 1$. That objective is strictly concave and the constraint set is convex, so a local maximiser is the unique global one, which is $\mathcal{E} = r$. Hence if r does not select $d^\star$, no strictly selecting maximiser exists. □

In words, the most informative profile that leads to the target is one of two things. If the likelihood ratio already leads to the target, it is the likelihood ratio. If not, it is a profile under which the target only just wins, tied with a competitor. Informativeness pulls the profile towards the likelihood ratio, and the constraint stops it at the point where the target would stop winning.

This is useful for two reasons. It shows the price of reaching a target that the evidence does not favour: the resulting profile never prefers the target by a margin, only through a tie. And it makes the problem easy to solve, since only the profiles where the target ties with a competitor need to be searched, which in the example below means solving two equations.

Nothing in the argument depends on the logarithm. Any information measure that is strictly concave and increasing in every coordinate behaves the same way, with r replaced by that measure's own optimum. Two practical points follow. If the target must win strictly, the best profile can be approached but not reached, although a profile just inside the boundary loses almost no information. And for a sensible target the constraint can always be met, since by Corollary 3.2 the safe profile meets it.

Return to the two-outcome example with the bet d_1 as target, and write $e_A = \mathcal{E}(A)$ and $e_B = \mathcal{E}(B)$. Validity under P_0 and the condition for d_1 to be selected read

$$\tfrac{1}{2}e_A + \tfrac{1}{2}e_B \le 1, \quad \text{that is} \quad e_A + e_B \le 2, \qquad \text{and} \qquad \frac{8}{e_B} \le \frac{4}{e_A}, \quad \text{that is} \quad e_B \ge 2e_A.$$

The likelihood ratio $(1.5, 0.5)$ fails this, as seen above. Since more e-value always raises expected log-evidence, the best profile spends the whole budget and sits exactly on the edge where d_1 stops winning, so $e_A + e_B = 2$ and $e_B = 2e_A$. Solving gives $3e_A = 2$, hence

$$\mathcal{E}^{\text{info}} = (2/3, 4/3), \qquad C(d_1; \mathcal{E}^{\text{info}}) = \frac{8}{4/3} = 6.$$

Compared with the safe profile $(0, 2)$, whose certificate is 4, the informative profile gives up two units of certificate. In return it has a finite log score,

$$0.75 \log \tfrac{2}{3} + 0.25 \log \tfrac{4}{3} = -0.232,$$

while the safe profile scores $-\infty$ because it puts no e-value on A, the outcome the belief considers most likely. Every later log score is computed the same way.

3.3 The Protection–Information Frontier

The last two results give two profiles for the same target. The safe profile gives the strongest guarantee but ignores the belief. The informative profile gives the most information the target allows, at the price of a weaker guarantee. A decision-maker who wants some of each can choose a profile between them. In the two-outcome example this whole range can be written down.

Every profile worth considering spends the whole budget, since unused e-value could raise a score and improve both the guarantee and the information. In the example this means $e_A + e_B = 2$, so a single number fixes the profile. Write

$$e_A = 2t, \qquad e_B = 2 - 2t, \qquad 0 < t \leq 1/3, \tag{3.2}$$

where t is the share of the budget placed on A. The bet is selected when $e_B \geq 2e_A$, which here reads $2 - 2t \geq 4t$, that is $t \leq 1/3$, and it is selected strictly when $t < 1/3$. As t approaches zero almost everything goes on B, and the profile approaches the safe profile. At $t = 1/3$ the bet only just wins, which is the informative profile. Beyond $1/3$ the hedge wins instead.

Moving t through this range trades one criterion for the other. The certificate

$$C(d_1) = \frac{8}{2 - 2t} = \frac{4}{1 - t}$$

rises from 4 to 6, so the guarantee weakens, while the expected log-evidence

$$0.75 \log(2t) + 0.25 \log(2 - 2t)$$

also rises, since its derivative $0.75/t - 0.25/(1 - t)$ is positive whenever $t < 3/4$, so the profile carries more information. This is why the most informative profile sits at the end of the range, $t = 1/3$, as found in Section 3.2. Smaller t buys protection, larger t buys information, and every t in the range still leads to the bet. Figure 1 shows the same range geometrically: panel (a) shows the valid profiles and the line $e_B = 2e_A$ where the two actions tie, and panel (b) the two certificates along the budget line, crossing at $t = 1/3$.

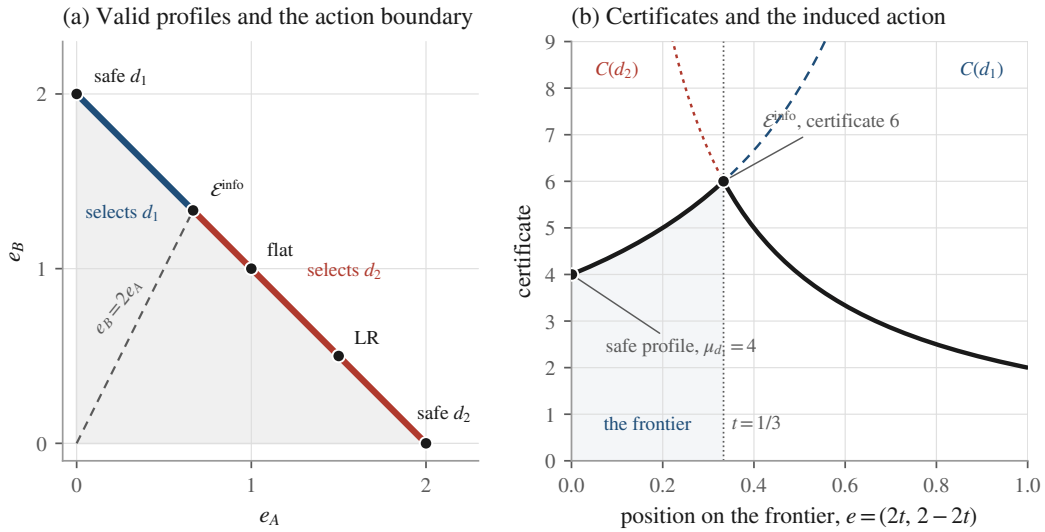


Figure 1. Geometry of the two-outcome problem. Panel (a) shows the valid profiles and the action boundary, and panel (b) the two certificates along the budget line.

Plotting the guarantee against the information gives the frontier in Figure 2. It is curved, so the exchange rate between the two is not constant: the first units of certificate given up buy a great deal of information, and the last buy very little. Most of the available information can therefore be had for a modest loss of protection. The protective endpoint itself is a limit of the curve rather than a point on it, since the safe profile puts no e-value on A and its log score is $-\infty$.

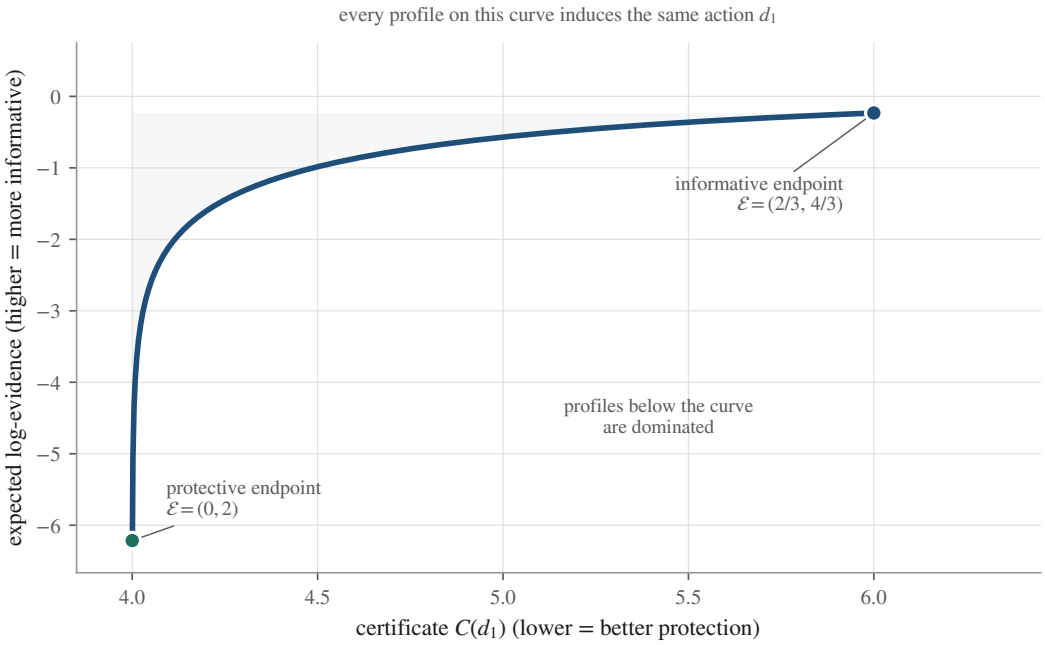


Figure 2. The protection–information frontier for profiles inducing the bet on A.

The same picture shows both cases of Proposition 3.3. Were the target the hedge d_2, the likelihood ratio $(1.5, 0.5)$, which sits at $t = 3/4$, would already lead to it, so the most

informative profile for d_2 would be the ratio itself, with d_2 winning strictly at certificate $8/3$. For the bet d_1 the ratio is out of reach, so the most informative profile stops at the tie $t = 1/3$. Which case applies depends only on whether the likelihood ratio already leads to the target.

Where on the frontier to sit is a matter of preference, which the loss table alone does not settle.

3.4 Result 3: When Shape Functions Suffice

The first two results build profiles directly from the loss. The standard construction never looks at the loss. It takes the likelihood ratio r and passes it through a shape function w, reporting $\mathcal{E}(z) = w(r(z))$. Could a well-chosen shape function have produced the safe profile anyway? If so, the loss would matter only for choosing w, and existing theory would already cover the design problem.

There is a simple reason to doubt it. A shape function sees each outcome only through its likelihood ratio, so two outcomes with the same ratio receive the same score, however differently they cost. The safe profile scores outcomes by their cost, so whenever two outcomes share a ratio but not a loss, no shape function can reproduce it.

The shape functions used in practice come from the capped power family of Koning and van Meer [2025],

$$w_h(r) = \min\left(\frac{r^{1/(1-h)}}{\lambda},\ \frac{1}{\alpha}\right). \tag{3.3}$$

It raises the ratio to a power, rescales it by the constant λ that makes the profile valid, and caps every score at $1/\alpha$. The parameter $h < 1$ sets how aggressively evidence is accumulated: $h = 0$ returns the ratio itself, and lower values flatten the profile. Different statisticians prefer different h and α, but every member of the family reads each outcome only through its ratio. The next result says exactly when that is enough.

Proposition 3.4 (When shape functions suffice)**.** *Assume $\mu_d > 0$, and let $r(z) := \widehat{P}_1(z)/P(z)$ be the likelihood ratio against any fixed $P \in \mathcal{P}$ giving every outcome positive probability. The safe profile $\mathcal{E}_d^{\text{safe}}$ of Theorem 3.1 can be written as $\mathcal{E}_d^{\text{safe}} = w(r)$ for some function w exactly when the loss is determined by the likelihood ratio, meaning that*

$$r(z) = r(z') \quad \textit{implies} \quad L(d, z) = L(d, z')$$

for every pair of outcomes z, z'.

Proof. Suppose first that $\mathcal{E}_d^{\text{safe}} = w(r)$ for some w. If $r(z) = r(z')$ then $w(r(z)) = w(r(z'))$, so the two outcomes receive the same e-value. Multiplying by μ_d recovers the loss, since $L(d, \cdot) = \mu_d\, \mathcal{E}_d^{\text{safe}}$, hence $L(d, z) = L(d, z')$.

Conversely, suppose the loss is constant on each level set of r. Define w on the values r actually takes by picking, for each such value v, any outcome z with $r(z) = v$ and setting

$$w(v) = \frac{L(d, z)}{\mu_d}.$$

The choice of z does not matter: any other outcome with the same ratio carries the same loss by assumption, so w is well defined. Off the range of r it may be given any values at all. By construction, at every outcome,

$$w(r(z)) = \frac{L(d, z)}{\mu_d} = \mathcal{E}_d^{\text{safe}}(z).$$

□

In words, a shape function can reproduce the safe profile exactly when outcomes with the same likelihood ratio always carry the same loss. This can fail in two different ways, and they should be kept apart.

The first is a failure of information. When two outcomes share a ratio but differ in loss, nothing computed from the ratio can tell them apart, so no shape function reproduces the safe profile, however it is chosen.

The second is subtler. When every outcome has its own ratio, the proposition allows a suitable shape function to exist, but that function is only the loss rewritten as a function of r after the fact. A statistician choosing w from preferences about evidence has no reason to arrive at it. Moreover, the shape functions used in practice are increasing in r, so they give their largest scores to the outcomes the belief rates likeliest relative to the model, while the safe profile gives its largest scores to the outcomes where the target is most expensive. When these are different outcomes, no increasing shape can follow the safe profile, and Section 4.4 shows a case where this alone puts the target out of reach.

3.5 The Procedure, Worked Through

The three results combine into a procedure with three steps. First, compare the actions and name the target. Second, build the safe profile for the target. Third, if information matters, trade some protection for it while keeping the target as a constraint. Each step is carried out below on the two-outcome example, so every number can be checked by hand.

Step one compares the actions in the two ways a decision-maker might care about, by worst-case expected loss across the model and by expected loss under the belief. In the example,

$$\max_{P \in \mathcal{P}} \mathbb{E}_P[L(\cdot, Z)] = \left(\tfrac{0+8}{2}, \tfrac{4+0}{2}\right) = (4, 2), \qquad \mathbb{E}_{\widehat{P}_1}[L(\cdot, Z)] = (2, 3),$$

so the protective action is the hedge d_2, while the belief recommends the bet d_1. The two differ, so the decision-maker has to choose. Take the bet d_1 as the target, the choice of a belief-trusting decision-maker. Everything that follows is conditional on that choice.

Step two builds the safe profile $L(d^\star, \cdot)/\mu_{d^\star}$. By Theorem 3.1 it gives the target the strongest guarantee, and by Corollary 3.2 it leads to the target. For d_1, $\mu_{d_1} = 4$ and $\mathcal{E}^{\text{safe}}_{d_1} = (0, 8)/4 = (0, 2)$, giving

$$C(d_1) = \max(0/0,\ 8/2) = 4, \qquad C(d_2) = \max(4/0,\ 0/2) = \infty,$$

under the conventions $0/0 = 0$ and $x/0 = \infty$. The bet is selected. The infinite certificate of the hedge is not an artefact. The safe profile puts no e-value on A, the only outcome in which hedging costs anything, so that loss is divided by zero rather than reduced.

Step three is optional. If the decision-maker also wants information, the statistician moves along the frontier of Section 3.3, keeping the target as a constraint. The most informative end is $\mathcal{E}^{\text{info}} = (2/3, 4/3)$, with certificate 6 instead of 4 and log score -0.232 instead of $-\infty$.

Table 1 collects the profiles. The two an evidence-first statistician would report, the likelihood ratio and the flat profile, both lead to the hedge. Only the profiles built from the loss of d_1 lead to the bet. Each safe profile achieves its own action's best certificate and sends the other action's to infinity, for the reason just seen.

Table 1. Two-outcome profiles. C_j is the certificate of action d_j.

Profile	$\mathcal{E}(A)$	$\mathcal{E}(B)$	C_1	C_2	Action
Likelihood ratio	1.50	0.50	16.00	2.67	d_2
Flat	1.00	1.00	8.00	4.00	d_2
Safe for d_2	2.00	0.00	∞	2.00	d_2
Safe for d_1	0.00	2.00	4.00	∞	d_1
Informative for d_1	0.67	1.33	6.00	6.00	d_1

The last row is the informative profile. Its two certificates are equal, so d_1 wins only through a tie, as Proposition 3.3 predicts when the likelihood ratio leads elsewhere. The three-outcome example behaves the same way. Targeting d_A there, the most informative profile is exactly

$$\mathcal{E} = (\tfrac{9}{5},\ \tfrac{12}{35},\ \tfrac{6}{7}) \approx (1.80,\ 0.34,\ 0.86),$$

with log score $+0.016$, and all three certificates equal $35/3 \approx 11.67$, so d_A ties with both competitors. A hand-picked profile that leans towards the draw, $(0.3, 0.7, 2.0)$, also leads to d_A, and strictly, but scores -0.760. Solving the constrained problem is therefore worth about 0.78 in log-evidence.

Every step of the procedure reads the loss. The likelihood ratio is still worth reporting as a measure of evidence, but it is not a default the decision should inherit.

4 Application: Stocking Shirts for the World Cup Final

Every example so far has stipulated its distributions. This section applies the procedure of Section 3.5 to real data, taking the belief from Polymarket prices for the 2026 FIFA World Cup final between Spain and Argentina [Marketlens, 2026]. The decision-maker is a shopkeeper near the stadium who orders replica shirts before the final and cannot reorder afterwards. Shirts of the winning side sell and the rest are marked down. The shopkeeper does not bet on the match and uses the prices only as a forecast, so the cost of an order depends on how well it matches demand, never on the prices themselves.

The section reaches one finding. Under the market's forecast the cheapest order is mostly Spain shirts, yet no increasing reshaping of the likelihood ratio leads to that order, across two extensive searches, while the safe profile does. The reason is where each places its evidence. Protecting a Spain-heavy order needs large scores on the Argentina outcomes, where that order is ruinous. A reshaped likelihood ratio puts its largest score on a Spain win within ninety minutes, where the market is most confident and the order costs almost nothing. This is Result 2 in real data: the profile that best reflects the belief leads away from the action the belief recommends.

4.1 Two Contracts, Four Outcomes

Two contracts were traded on the final, each settling on a different question. A three-way result market pays out on the score after ninety minutes, so a match still level then settles as a draw. A binary trophy market pays out on whichever side eventually lifts the cup. Together they tell the shopkeeper which side wins and whether the match is decided within ninety minutes, and both affect how many shirts sell.

Figure 3 shows the two on one timeline. Prices are recorded every minute of clock time, so breaks in play appear as gaps, which is why the goal in the 106th minute of play falls more than two hours after kick-off. The two series part at the end of regulation. With the match level the draw contract settles and the result market stops moving, while the trophy market stays live and Spain's extra-time goal lifts their price from about 0.60 to over 0.90. These are traded prices, not e-values.

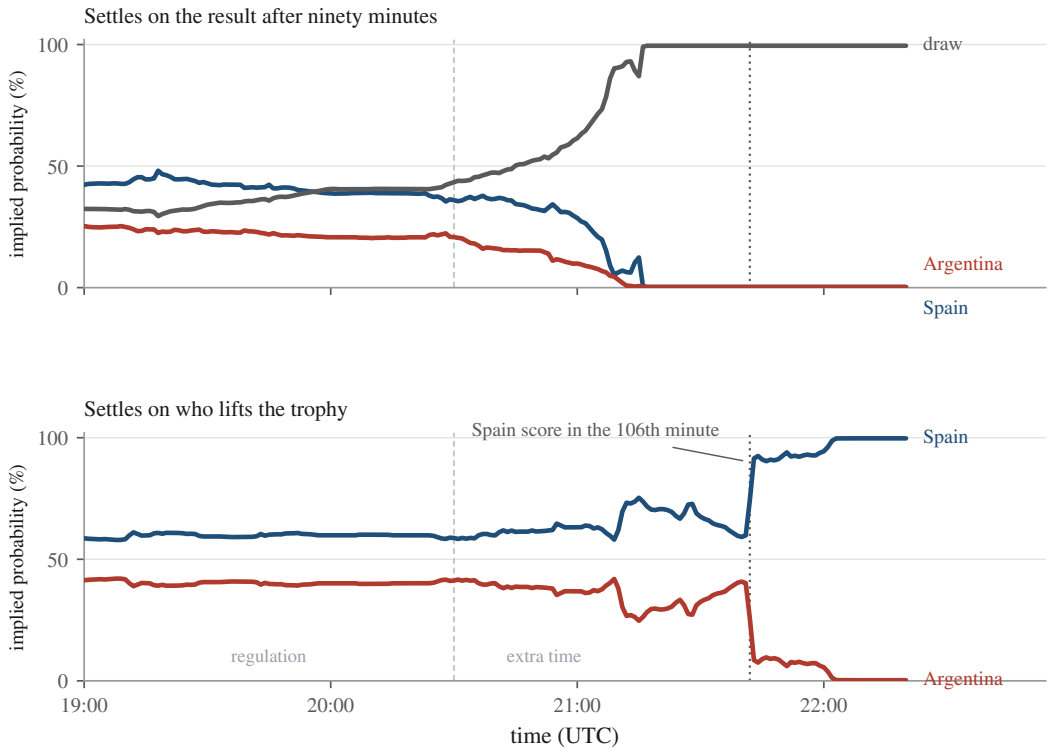


Figure 3. Two Polymarket contracts on the same match, from kick-off to settlement. The dashed line marks the end of regulation, the dotted line the goal.

At kick-off the result market quoted

$$q = (0.4231,\ 0.3238,\ 0.2531)$$

for a Spain win, a draw and an Argentina win, and the trophy market quoted $o = (0.5863, 0.4137)$ for Spain and Argentina. Each number is the midpoint of the best bid and ask. Midpoints of separately traded contracts need not sum to one, so each set is rescaled to sum to one, here and in Figure 3. The data also record how both contracts settled: the draw contract paid out, and the trophy went to Spain [Marketlens, 2026].

Read together, the two contracts describe four outcomes,

$$\mathcal{Z} = \{s,\ s^{\mathrm{e}},\ a,\ a^{\mathrm{e}}\},$$

where s is a Spain win within ninety minutes, s^{e} a Spain win after extra time or penalties, and a and a^{e} the same for Argentina. Four outcomes need three numbers, and the two contracts supply exactly three. The result market gives the two wins within ninety minutes directly, and subtracting these from the trophy prices leaves the probability of each side winning later:

$$\widehat{P}_1(s) = q_1, \quad \widehat{P}_1(s^{\mathrm{e}}) = o_1 - q_1, \quad \widehat{P}_1(a) = q_3, \quad \widehat{P}_1(a^{\mathrm{e}}) = o_2 - q_3. \tag{4.1}$$

A match level after ninety minutes must eventually be won by someone, so the two late wins should together account for the draw price, and they do: $0.1632 + 0.1606 = 0.3238 = q_2$. This holds by construction, since both sets of prices were rescaled to sum to one, so it checks the arithmetic rather than the market. The contracts cannot tell extra time from penalties, and nothing below depends on that distinction.

These prices give the belief only. The model, the set of distributions the guarantee must hold under, is not read from the prices at all. It is built separately in the next subsection.

4.2 The Model and the Belief

The procedure needs a belief and a model, and they do different jobs. The belief is the shopkeeper's best guess at what will happen, and it decides which order looks cheapest. The model is the set of distributions the guarantee must hold up against, and it decides how strong a guarantee any profile can give. Both must be fixed before kick-off, when the order is placed.

The belief is the market's view at kick-off. Applying (4.1) to the opening quotes gives

$$\widehat{P}_1 = (0.4231,\ 0.1632,\ 0.2531,\ 0.1606).$$

Later prices cannot replace it. They absorb the match as it is played, and by the end nearly all their probability sits on the outcome that happened, so they report the result rather than forecast it.

The model needs more explanation. The guarantee is meant to hold even if the market is wrong about this match, so the model cannot simply be the market's forecast, or the guarantee would be only as good as the market. Instead it collects a few distributions built from information available before the two finalists were known, and the guarantee must hold under all of them. These are not meant to be better forecasts than the market, which is better informed. They are the alternatives the shopkeeper wants to be protected against.

Each member splits the probability between the two sides, and then splits each side between winning within ninety minutes and winning later. Three splits between the sides are used: Polymarket's tournament-winner prices early in the group stage, 0.138 for Spain and 0.116 for Argentina [Polymarket, 2026], rescaled to sum to one as $(0.5433, 0.4567)$, an even split, and $(0.45, 0.55)$, leaning to Argentina. The share of matches decided later comes from history. Of the twenty-one World Cup finals played as a single deciding match between 1930 and 2022, eight went beyond ninety minutes [Wikipedia contributors, 2026], a rate of $8/21 = 0.381$. The 1950 tournament is excluded because it ended with a final group stage rather than a final. For the first split this gives

$$\begin{aligned} P &= \big(0.5433\,(0.619),\ 0.5433\,(0.381),\ 0.4567\,(0.619),\ 0.4567\,(0.381)\big) \\ &= (0.3363,\ 0.2070,\ 0.2827,\ 0.1740), \end{aligned}$$

and the other two follow in the same way, so

$$\begin{aligned}\mathcal{P} = \{&(0.3363,\ 0.2070,\ 0.2827,\ 0.1740),\\ &(0.3095,\ 0.1905,\ 0.3095,\ 0.1905),\\ &(0.2786,\ 0.1714,\ 0.3405,\ 0.2095)\}.\end{aligned}$$

The model is deliberately more cautious than the belief. The market puts 0.4231 on a Spain win within ninety minutes, while no member of the model goes above 0.3363. By Theorem 3.1, only one member matters for a given order, the one under which that order looks worst, and Section 4.4 checks that the findings survive changes to the historical rate.

4.3 The Two Criteria Disagree

The shopkeeper has room for one hundred shirts and must split them between the two sides. Three orders are considered: mostly Spain (d_1, ninety Spain shirts), an even split (d_2, fifty) and mostly Argentina (d_3, twenty). Each shirt short of demand costs the lost margin, three units, and each shirt left unsold costs the markdown, one unit.

Demand depends on the outcome. Only the winning side's shirts sell well, and a win after extra time is a bigger occasion, so it sells more. Demand for the winner's shirts is taken as eighty after a win within ninety minutes and one hundred after a later win, with ten of the losing side either way. The cost of each order then follows by arithmetic. In tens of units, for the outcomes $(s, s^{\mathrm{e}}, a, a^{\mathrm{e}})$,

$$L(d_1, \cdot) = (1, 3, 29, 35), \quad L(d_2, \cdot) = (13, 19, 13, 19), \quad L(d_3, \cdot) = (25, 31, 1, 7).$$

For example, ordering mostly Spain when Spain win within ninety minutes leaves ten Spain shirts unsold, a cost of ten units, or 1 in the table. Ordering mostly Spain is cheap when Spain win and ruinous when Argentina do. The even split is never cheap and never ruinous, and its cost depends only on when the match is decided, not on who wins. No price appears anywhere in the table: the market says what is likely, and the shirts say what it costs to be wrong.

Step one compares the orders in the two ways of Section 3.5, by worst-case expected loss across the model and by expected loss under the belief,

$$\begin{aligned}\max_{P \in \mathcal{P}} \mathbb{E}_P[L(\cdot, Z)] &= (18.0000,\ 15.2857,\ 16.3251),\\ \mathbb{E}_{\widehat{P}_1}[L(\cdot, Z)] &= (13.8738,\ 14.9430,\ 17.0142).\end{aligned}$$

The two disagree. The protective action is the even split d_2, the order that cannot go badly wrong. The belief recommends mostly Spain, d_1, because the market makes Spain the likelier winner. This is the case the thesis is about.

Step two takes d_1 as the target, the choice of a belief-trusting shopkeeper. The worst member of the model for d_1 is the one leaning to Argentina,

$$P^\star = (0.2786,\ 0.1714,\ 0.3405,\ 0.2095),$$

under which

$$\mu_{d_1} = 1\,(0.2786) + 3\,(0.1714) + 29\,(0.3405) + 35\,(0.2095) = 18.0000,$$

so the safe profile of Theorem 3.1 divides the losses of d_1 by 18,

$$\mathcal{E}^{\text{safe}}_{d_1} = \frac{(1, 3, 29, 35)}{18} = (0.0556,\ 0.1667,\ 1.6111,\ 1.9444). \tag{4.2}$$

It puts most of its e-value on the two Argentina outcomes, where d_1 is most exposed. Its certificate is 18, and since d_1 is the action the belief recommends, Corollary 3.2 guarantees that it leads to d_1. Table 2 compares it with the profiles an evidence-first statistician would report. The likelihood ratio there is taken against $P^\star$ and scaled down by a factor of 1.051, so that it is valid under every member of the model.

Table 2. The shopkeeper's order. Certificates C_j of each order under each profile, the order the rule selects, and that order's expected loss under the kick-off belief.

Profile	C_1	C_2	C_3	Action	$\mathbb{E}_{\widehat{P}_1}[L]$
Likelihood ratio	48.002	26.058	34.233	d_2	14.943
Flat	35.000	19.000	31.000	d_2	14.943
Safe for d_2	34.099	15.286	29.396	d_2	14.943
Safe for d_1	18.000	234.000	450.000	d_1	13.874

Only the safe profile for d_1 leads to d_1. The likelihood ratio and the flat profile both leave the shopkeeper with the even split, and so does the safe profile for d_2. This is the failure Section 3.2 warned of: the likelihood ratio does not lead to the action the belief itself recommends. Measured under the belief, that costs the shopkeeper about 1.07 in expected loss, 14.943 against 13.874. The prediction-set rule of Section 1.1 does no better: at every coverage level from 50% to 95% it also chooses the even split.

4.4 No Increasing Shape Reaches the Target

Only the safe profile led to the order the belief recommends. A natural objection is that the standard construction was not tried hard enough, and that some other shape function would have led to d_1 as well. This subsection shows that none does.

The standard construction reshapes the likelihood ratio of the belief against the member of the model under which validity binds, here the worst member $P^\star$, so

$$r = \frac{\widehat{P}_1}{P^\star} = (1.5188,\ 0.9521,\ 0.7433,\ 0.7666).$$

The ratio takes a different value at every outcome, so Proposition 3.4 does not rule out writing the safe profile as $w(r)$.

The obstacle is the second one described after that proposition. The shape functions used in practice are increasing in r, so they rank the outcomes as r does,

$$a \prec a^{\mathrm{e}} \prec s^{\mathrm{e}} \prec s,$$

giving the most e-value to s, the outcome the belief rates likeliest relative to the model. The loss of d_1 ranks them almost the other way,

$$s \prec s^{\mathrm{e}} \prec a \prec a^{\mathrm{e}}.$$

To keep the certificate of d_1 low, the profile must put its largest e-value on the two Argentina outcomes, where ordering mostly Spain is ruinous. An increasing shape does the opposite. It puts its largest e-value on s, where the order costs almost nothing, and its smallest on a and a^{e}, where the order is most exposed. The safe profile (4.2) reverses this, which is why it is not increasing in r.

Failing to match the safe profile exactly is not the same as failing to reach d_1, so the question is settled by searching over shapes directly. Two searches are run, both keeping every profile valid with its budget fully spent, so that only the shape varies. The first covers the capped power family (3.3) of Section 3.4, sweeping 120 values of h in $[-6, 0.95]$ against 60 values of α in $[0.01, 0.95]$, a total of 7,200 combinations, with λ set by bisection. The second drops the parametric form altogether. It draws 200,000 profiles by sampling four values uniformly, sorting them and assigning them to the outcomes in increasing order of r, which gives a random increasing shape.

Neither search ever reaches d_1. Every shape tried leads to the even split or to mostly Argentina. Restricting to increasing shapes therefore does not merely make d_1 more expensive to protect. It puts d_1 out of reach altogether, while the safe profile reaches it with certificate 18.

One input is left to the analyst, the historical rate at which finals go beyond ninety minutes. Repeating the analysis for rates from 6/21 to 10/21, two finals either side of the observed 8/21, the two criteria disagree at all five rates, and at all five no capped power shape reaches d_1. The finding does not depend on that choice.

What the application cannot show is which criterion was right. Spain won after extra time [Al Jazeera, 2026], the outcome s^{e}, in which ordering mostly Spain costs 3 and the even split 19. But a single match is not a verdict on a decision rule. Settling that requires a setting where the truth is known in advance.

5 Controlled Simulations: Scoring the Decisions Against a Known Truth

The application showed the two criteria disagreeing on real data, but a single match cannot tell which of them was right. It produced one outcome, and an order that was cheap this time could still be the worse choice on average. Ranking decision rules needs the distribution that generates the outcomes, which no real match supplies. This section therefore runs controlled simulations, in which that distribution is fixed in advance and every rule can be scored against it.

Each simulation adds a third distribution to the model and the belief, the truth P_{true}. It plays no part in choosing the action. Once an action is chosen, it is scored by its expected loss under the truth,

$$\mathbb{E}_{P_{\text{true}}}\big[L(\delta(\mathcal{E}), Z)\big].$$

The truth need not lie in the model, since the model is what the decision-maker wants protection against, not a claim about where the truth lies.

The simulations compare four ways of choosing the evidence. Two follow the procedure of Section 3.5, one targeting the action the belief recommends and one targeting the protective action, and by Corollary 3.2 each takes its target. The other two are the likelihood ratio and the flat profile, which take whatever action the rule selects under them. The first simulation varies how confident the belief is, the second how much worse a losing bet is than an unnecessary hedge, and the third how wrong the belief is. Since every distribution is known, each expected loss is exact.

5.1 The Likelihood Ratio Moves Against the Belief

The first simulation asks how each rule responds as the belief grows more confident. Keep the two-outcome loss, with the bet $L(d_1, \cdot) = (0, 8)$ and the hedge $L(d_2, \cdot) = (4, 0)$, hold the truth at $(0.75, 0.25)$, and let the belief put probability p on A.

The belief's target is the bet when betting is cheaper under the belief, $8(1-p) \le 4p$, that is $p \ge 2/3$. More confidence in A makes it readier to bet, as one would expect. The likelihood ratio against $P_0 = (1/2, 1/2)$ is $\mathcal{E} = (2p, 2(1-p))$, with certificates

$$C(d_1; \mathcal{E}) = \frac{4}{1-p}, \qquad C(d_2; \mathcal{E}) = \frac{2}{p},$$

so it bets only when $p \le 1/3$. The two thresholds point in opposite directions. The likelihood ratio bets on A only when the belief makes A unlikely, and above $p = 2/3$, where betting is genuinely cheaper, it hedges, more firmly the more confident the belief. Figure 4 shows the actions each rule takes and their expected loss under the truth.

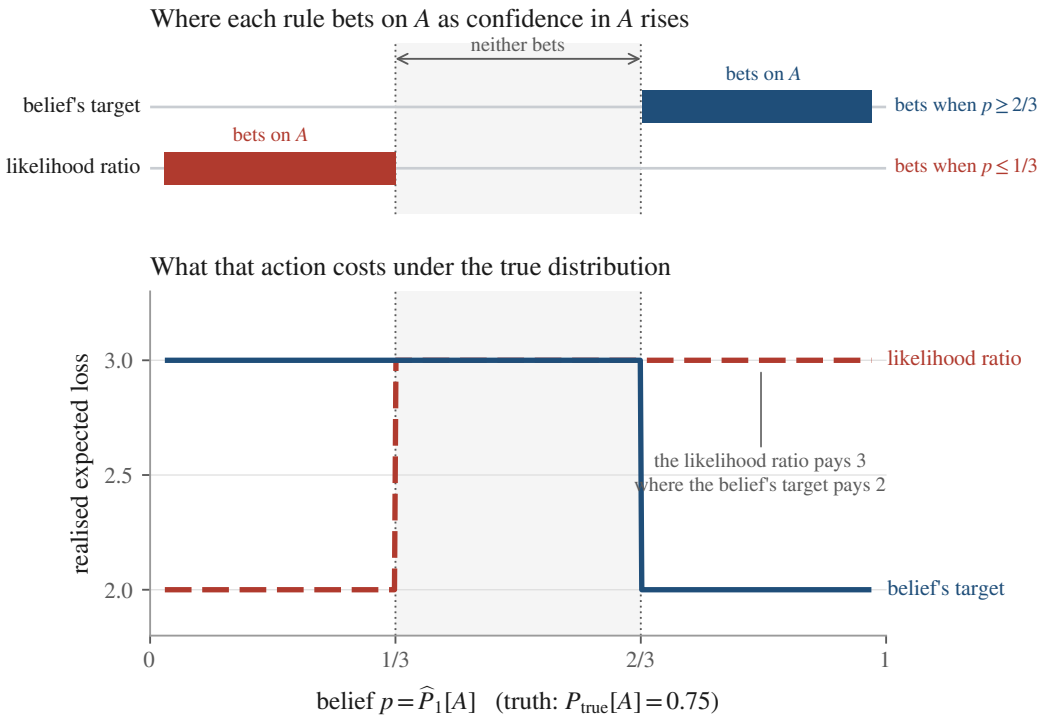


Figure 4. Actions taken and expected loss under the truth as confidence in A varies. The procedure targeting the protective action and the flat profile hedge at every p, so only the two rules that ever bet are drawn.

This is the mechanism of Section 3.2 again. The likelihood ratio puts large e-values on the outcomes the belief favours, so the rule discounts the losses there. The bet loses only in B, which the belief disfavours, so its loss is left undiscounted. That suits a profile meant as evidence against the model, not one meant to act on a belief.

5.2 How Wide the Disagreement Is

The second simulation varies the loss instead. Set the belief and the truth both to $(0.75, 0.25)$, so the belief is correct, and write ℓ for the cost of a losing bet and k for the cost of an unnecessary hedge, 8 and 4 in the example. With $p = 0.75$, the bet is cheaper under the belief when $\ell/k \le p/(1-p) = 3$, but the likelihood ratio selects it only when $\ell/k \le (1-p)/p = 1/3$. In between, the likelihood ratio hedges although betting is cheaper,

$$\frac{1-p}{p} < \frac{\ell}{k} < \frac{p}{1-p}, \qquad \text{here} \qquad \frac{1}{3} < \frac{\ell}{k} < 3. \tag{5.1}$$

The example's ratio of 2 lies inside this window, which is why it separates the two rules.

Since the belief is correct, the procedure targeting the belief's action has the smallest expected loss at every ratio. What the simulation measures is the width of the window, and (5.1) shows that it widens as the belief grows more confident. Figure 5 plots the expected losses and shades the window.

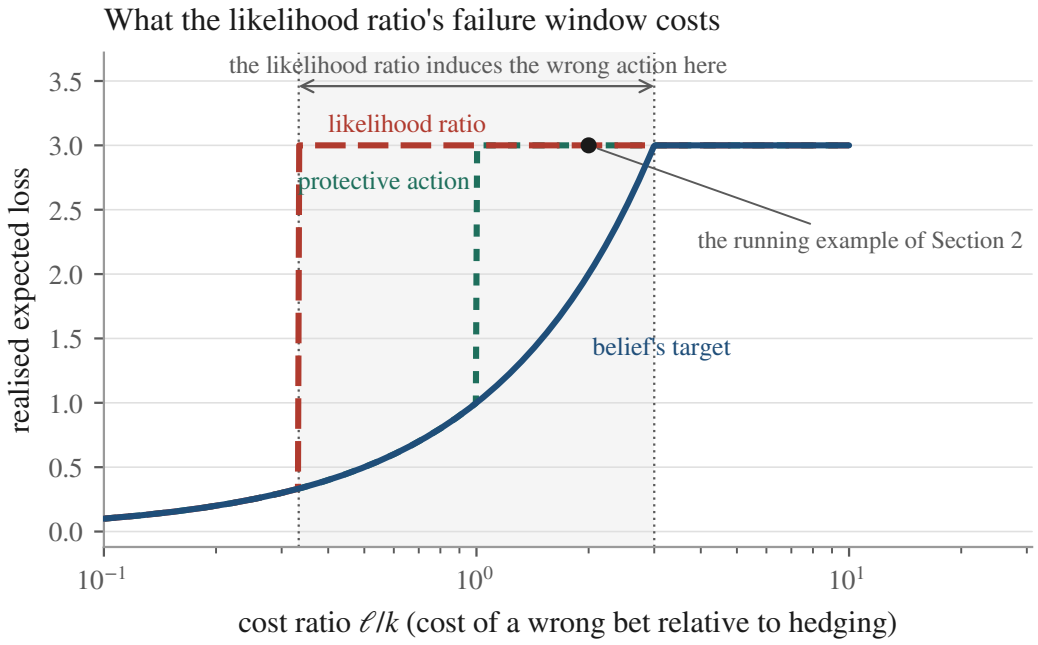


Figure 5. Expected loss against the cost ratio ℓ/k. Shading marks the window (5.1) in which the likelihood ratio takes the wrong action.

5.3 How Far an Optimistic Belief Can Be Trusted

The first two simulations concern the profile: given a target, which action does each profile lead to? This one concerns the target itself, which Section 3 leaves to the decision-maker's attitude toward risk. It makes the belief wrong and asks how wrong it can be before trusting it stops paying.

Take the three-outcome example of Section 2.4, where the belief $(0.6, 0.3, 0.1)$ puts only 0.1 on the draw. Let the truth be

$$P_{\text{true}} = \left(\tfrac{2}{3}(1-c),\ \tfrac{1}{3}(1-c),\ c\right),$$

and raise the true probability c of a draw from 0.01 to 0.45. The belief-trusting procedure takes d_A throughout, since the belief never changes, while the guarantee-first procedure, the likelihood ratio and the flat profile all take the insurance d_C. The expected losses of d_A and d_C under the truth are equal at $c^* = 1/4$. Below it trusting the belief costs less, and above it insuring does.

The crossover comes well before the belief stops ranking the outcomes correctly, since A remains the most likely outcome until $c = 2/5$. Ranking correctly is not enough when the outcome the belief dismisses costs 10. The threshold comes from the loss table and the truth together. Draws are common in football, so a belief that puts only 0.1 on them may well be on the wrong side of it.

The simulation does not say which target to name, since that depends on how far the decision-maker trusts the belief. What it shows is what each choice costs if the belief is wrong, which is exactly what a stated attitude toward risk has to weigh.

Taken together, the simulations make a narrow point. The likelihood ratio can take the wrong action even when the belief is exactly right, and over a wide range of loss asymmetries. When the belief may be wrong, how far to trust it depends on the loss table and the truth together, not on whether the belief ranks the outcomes correctly.

What these simulations cannot do is show that the procedure decides better in general, since their distributions are constructed rather than observed. That is the price of knowing the truth at all, and Section 6.2 returns to it.

6 Conclusion

6.1 What the Thesis Shows

The loss must enter the construction of the evidence, not only its interpretation. For a named target, the profile proportional to its loss is the unique safest choice and leads to it. Information can be added only with the target held as a constraint, since the likelihood ratio can lead elsewhere even when the belief is right, and no reshaping of the ratio does this job unless the loss is a function of it. The shopkeeper application shows that this matters in practice.

When the model and the belief favour different actions, the disagreement should be reported, since which to follow is the decision-maker's call. Reporting the likelihood ratio alone, without the loss and the action it induces, commits silently to an objective the decision may not share.

6.2 Limitations

Everything here is proved for finitely many outcomes. Theorem 3.1 should extend to richer outcome spaces, but the thesis does not prove that, and the frontier depends more on finiteness: with many actions or a continuum of outcomes, both the existence of an optimum and the shape of the edges where the target ties with a competitor need further assumptions.

Proposition 3.3 also assumes that exactly one validity constraint binds at the optimum. This holds for the one-point models it is applied to, but with a composite model two constraints can bind at once, and the most informative profile is then a compromise between members rather than a ratio against any single one.

The information criterion is a choice, not a claim about what decision-makers want. A different measure would move the interior of the frontier but not the protective endpoint, so the target, and how much protection to trade for information, must still be named by the decision-maker.

The empirical work is conditional too. The simulations construct their distributions, so they show how the mechanism behaves rather than rank rules in general. The application uses a real belief, but its loss table and model are the analyst's choice. Section 4.4 varies the input that is hardest to defend, the historical rate, and the finding survives, but a different loss table could change it.

6.3 Open Questions

Three questions remain open. The first is to extend the frontier beyond the finite setting. The second is to find loss tables that are determined by the likelihood ratio in practice, and to measure what is lost when they are not. The third is how the model $\mathcal{P}$ should be chosen. It sets the strength of every guarantee here, yet in the application it was assembled from defensible judgements rather than derived from a principle.

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A Notation

Table 3. Notation used in the thesis.

Symbol	Meaning
$\mathcal{Z}$	Outcome space
$\mathcal{D}$	Action space
$\mathcal{P}$	Model: the distributions validity must hold under
P_0	The single member of a one-point model $\mathcal{P} = \{P_0\}$
$\widehat{P}_1$	Belief: the distribution the decision-maker thinks is true
P_{true}	Distribution generating outcomes in the simulations of Section 5
$P^\star$	Member of $\mathcal{P}$ making the target action look worst
$L(d, z)$	Loss of action d when outcome z occurs
$\mathcal{E}(z)$	E-value assigned to candidate outcome z
$r(z)$	Likelihood ratio $\widehat{P}_1(z)/P^\star(z)$
$w(\cdot)$	Shape function, giving $\mathcal{E}(z) = w(r(z))$
w_h	Capped power shape with curvature h and scale λ
S	Prediction set
$s, s^{\text{e}}, a, a^{\text{e}}$	Outcomes of the application: Spain or Argentina winning within ninety minutes, or later
t	Share of the budget placed on A along the two-outcome frontier
$C(d; \mathcal{E})$	Weighted worst-case certificate
μ_d	Worst expected loss $\max_{P \in \mathcal{P}} \mathbb{E}_P[L(d, Z)]$
$\mathcal{E}_d^{\text{safe}}$	Safe profile for d, $L(d, \cdot)/\mu_d$
$\mathcal{E}^{\text{info}}$	Constrained information-optimal profile
$\mathcal{E}^{\text{flat}}$	Flat profile, equal e-value at every outcome
$d^\star$	Target action the profile is designed to induce
$\delta(\mathcal{E})$	Action selected by the weighted worst-case rule
α	Level at which a profile is thresholded, $\mathcal{E} \geq 1/\alpha$, and the cap in w_h
ℓ, k	Cost of a losing bet and of hedging, in Section 5.2